

Matrix Theory: Lecture 9

Carl Olsson

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Outline

- Hermitian Conjugation
- Inner products
- Unitary Matrices
- Hermitian Matrices and Diagonalization



Hermitian Conjugation

Def. The **Hermitian conjugate** matrix A^H is defined as $A^H = \overline{A}^T = \overline{A}^T$ (transpose and conjugate in any order)

Theorem 12.1

Let A, B be complex matrices of the same size then:

- $(A^H)^H = A$,
- $(cA)^H = \overline{c}A^H$,
- $(A + B)^H = A^H + B^H$,
- $(AB)^H = B^H A^H$.
- $(A_1 A_2 \cdots A_n)^H = \begin{pmatrix} A_1^H \\ \vdots \\ A_n^H \end{pmatrix}$ and $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^H = \begin{pmatrix} A^H & C^H \\ B^H & D^H \end{pmatrix}$
- If A is a square matrix then $\det(A^H) = \overline{\det A}$.



Hermitian Conjugation

A matrix A is called **Hermitian** if

$$A^H = A.$$

(Corresponds to symmetric matrix in $\mathbb{R}^{n \times n}$.)

A matrix A is called **skew-Hermitian** if

$$A^H = -A.$$

(Corresponds to skew-symmetric matrix in $\mathbb{R}^{n \times n}$.)

A matrix A is called **unitary** if

$$A^H = A^{-1}$$

(Corresponds to orthogonal matrix in $\mathbb{R}^{n \times n}$.)



Inner Products

V vector space over $\mathbb{C}$.

Def. An **inner (or scalar) product**, $(\cdot, \cdot)$, is a function in two vector variables, which calculates a complex number (v, u) fulfilling:

- $(v, v) = 0 \Leftrightarrow v = 0$.
- (v, v) is a positive real number for $v \neq 0$.
- $(v, u) = \overline{(u, v)}$.
- $(v, \alpha u + \beta w) = \alpha(v, u) + \beta(v, w)$.

Ex. $V = \mathbb{C}^n$. For two vectors X, Y we have

$$(X, Y) = \overline{x_1}y_1 + \overline{x_2}y_2 + \cdots + \overline{x_n}y_n = \sum_i \overline{x_i}y_i = X^H Y.$$

and

$$(X, X) = \sum_i \overline{x_i}x_i = \sum_i |x_i|^2.$$



Unitary Matrices ($A^H = A^{-1}$)

Theorem 12.8

The following conditions are equivalent for a square matrix U :

- ① U is an **unitary matrix**. Thus it is invertible and $U^H = U^{-1}$,
- ② $U^H U = I$,
- ③ $U U^H = I$,
- ④ The columns form an orthonormal basis.
- ⑤ $(UX, UY) = (X, Y)$ for any column-vectors X, Y .
- ⑥ $|UX| = |X|$ for any column-vector X ,
- ⑦ $U^H U = U U^H$ and $|\lambda| = 1$ for all eigenvalues λ .
- ⑧ The condition number $\kappa(U) = 1$ and $|\det(U)| = 1$.



Unitary Matrices ($A^H = A^{-1}$)

Theorem 12.9

Let A, B be unitary matrices of the same size. Then

- $A^{-1} = A^H$ is also a unitary matrix,
- AB and AB^{-1} are unitary matrices,
- cA is a unitary matrix for $c \in \mathbb{C} \Leftrightarrow |c| = 1$,
- $|\det A| = 1$,
- $C = \begin{pmatrix} A & O \\ O & B \end{pmatrix}$ is unitary $\Leftrightarrow A$ and B unitary.
- If A is a triangular matrix then A is a diagonal matrix.



Gram-Schmidt

Theorem 12.6

For any basis $f_1, \dots, f_n$ there exists an orthonormal basis $e_1, \dots, e_n$, such that for any k the vectors $e_1, \dots, e_k$ span the same vector subspace as $f_1, \dots, f_k$ and the inner product (e_k, f_k) is a positive real number. In particular every orthonormal set of vectors can be completed to an orthonormal basis.

$$e_1 = \frac{f_1}{\sqrt{(f_1, f_1)}}.$$

$$v_{k+1} = f_{k+1} - \sum_{j=1}^k (e_j, f_{k+1}) e_j.$$

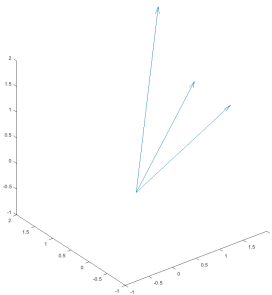
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Gram-Schmidt

Theorem 12.6

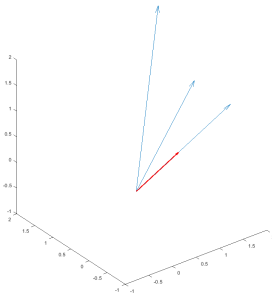
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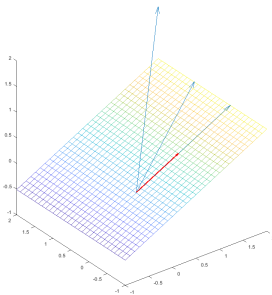
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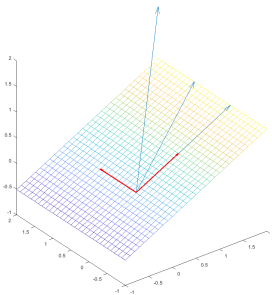
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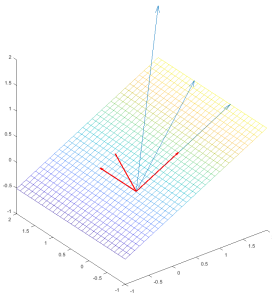
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QR-decomposition

Matrix formulation of Gram-Schmidt. f_1 lin. comb. of e_1 :

$$[f_1] = [e_1] [r_{11}]$$



QR-decomposition

Matrix formulation of Gram-Schmidt. f_2 lin. comb. of e_1, e_2 :

$$\begin{bmatrix} f_1 & f_2 \end{bmatrix} = \begin{bmatrix} e_1 & e_2 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} \\ 0 & r_{22} \end{bmatrix}$$



QR-decomposition

Matrix formulation of Gram-Schmidt. f_3 lin. comb. of e_1, e_2, e_3 :

$$\begin{bmatrix} f_1 & f_2 & f_3 \end{bmatrix} = \underbrace{\begin{bmatrix} e_1 & e_2 & e_3 \end{bmatrix}}_{\text{unitary}} \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ 0 & r_{22} & r_{23} \\ 0 & 0 & r_{33} \end{bmatrix}}_{\text{right triangular}}$$



QR-decomposition

Matrix formulation of Gram-Schmidt.

Theorem 12.11

Every invertible complex matrix A has a unique **QR-decomposition** (or QR-factorization), namely

$$A = UR,$$

where U is a unitary matrix and R is a upper triangular matrix with positive elements on the main diagonal (the elements above the diagonal can be complex). If A is a real matrix then both U and R are real, thus U is an orthogonal matrix.



The Adjoint Operator

Let $\mathcal{A}$ be a linear operator in a vector space V with an inner product. An operator $\mathcal{A}^*$ in the same vector space is called the **adjoint operator** if

$$(\mathcal{A}^* u, v) = (u, \mathcal{A} v)$$

for any pair of vectors u, v .

An operator $\mathcal{A}$ is **self-adjoint** if

$$(\mathcal{A} u, v) = (u, \mathcal{A} v)$$

for any pair of vectors u, v .



Hermitian matrices ($A^H = A$)

In coordinates: If $(X, Y) = X^H Y$ then

$$(X, AY) = X^H AY = (A^H X)^H Y = (A^H X, Y)$$

That is, A^H represents the adjoint of A .

The matrix representation of a self adjoint operator is **Hermitian** $A^H = A$.
(Generalization of symmetric $A^T = A$.)

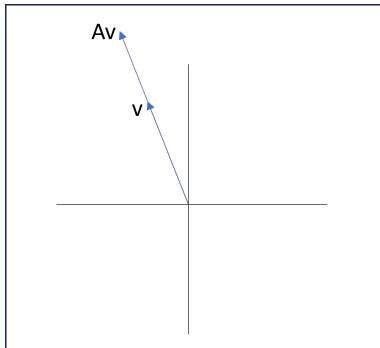
Theorem 12.15

Let $A = A^H$ be an Hermitian matrix. Then

- 1 All eigenvalues of A are real.
- 2 Eigenvectors with the different eigenvalues are orthogonal to each other.
- 3 There exists a unitary matrix U such that $U^H A U$ is a diagonal real matrix.

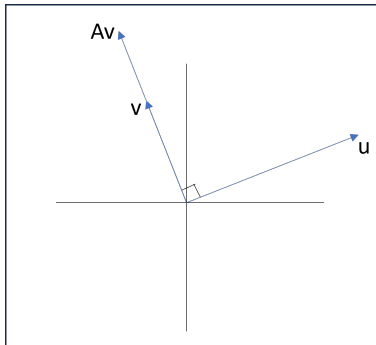
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Proof idea of 3) (real 2×2 case): Suppose we have one eigenvector:



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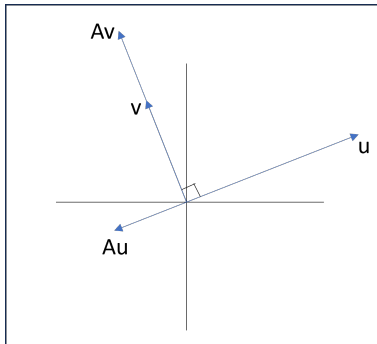


Consider subspace perpendicular to v . Does it contain an eigenvector?



Hermitian matrices ($A^H = A$)

Proof idea of 3) (real 2×2 case): Suppose we have one eigenvector:



Consider subspace perpendicular to v . Does it contain an eigenvector?

Yes, A maps perpendicular vectors to perpendicular vectors since

$$0 = \lambda(u, v) = (u, \lambda v) = (u, Av) = (Au, v)$$

Au must be parallel to u .

